# NAYAB KHALID

Data Science | Machine Learning | Artificial Intelligence
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## PROFILE

Final-year Bachelor of Science in Data Science student at COMSATS University Islamabad (CGPA 3.65/4.00) with academic and research experience in machine learning, computer vision, artificial intelligence, statistical analysis and data analytics. Experienced in developing and evaluating ML models through research studies, internships and multidisciplinary projects, including research on fairness in melanoma detection. Interested in postgraduate study in machine learning and data science, combining careful statistical evaluation with data-driven systems that perform robustly across diverse populations.

## EDUCATION

**COMSATS University Islamabad** **Feb 2023 – Jan 2027**
*Bachelor of Science in Data Science* | Current CGPA: 3.65 / 4.00 | Expected graduation: January 2027
**Relevant coursework:** Machine Learning, Artificial Intelligence, Data Mining, Statistical Methods in Data Science, Big Data Analytics, Parallel and Distributed Computing, Data Warehousing and Business Intelligence, Linear Algebra, Database Systems, Design and Analysis of Algorithms

**Punjab Group of Colleges, Ayub Park Campus** **Sep 2020 – Jul 2022**
*FSc, Pre-Engineering* | Grade: A1

**Saint Anne's Presentation Convent High School** **May 2017 – Feb 2020**
*Matric, Computer Science* | Grade: A+

## INTERNSHIPS & PROFESSIONAL EXPERIENCE

**CadetX** | *Data & AI Intern* **Aug 2026 – Nov 2026 (ongoing)**
- Building an NLP and forecasting pipeline over job-posting data to surface emerging skill trends and predict hiring demand.
- Collaborating with a four-person team in Python, coordinating development through GitHub to consolidate the analysis into a unified hiring intelligence system.

**FlyRank AI** | *Machine Learning Intern* **Jul 2026 – Aug 2026**
- Completed the Refresh and Content Opportunity Scoring capstone across 8 assignment notebooks: framed the research question, verified the data contract against a 9.8M-row warehouse release (30K pages, 44 columns, 32 clients), and built a transparent staleness and visibility baseline before modelling.
- Authored a detailed research paper based on the capstone, documenting the methodology, client-grouped validation strategy, and model performance improvements.

**Canteen Stores Department (CSD)** | *IT Intern* **Jul 2025 – Aug 2025**
- Analyzed COVID-19's impact on 1,000+ employee responses across 4 sectors in Python and Scikit-learn; compared 5 classifiers (Random Forest best on F1), forecast 6 months forward and delivered an interactive Tableau dashboard.
- Built OpenCV computer vision projects including Invisibility Cloak and Air Drawing.
- Studied CSD's data center end to end, covering infrastructure, systems operations and security, to understand how the organization's IT environment fits together.

## RESEARCH

**Domain Before Diversity: When Do Darker-Skin Training Images Improve Skin-Cancer Classification on Clinical Photographs?**
*Co-authored research study · Status: ongoing · with Noor Ul Huda Batool · 2026*
Ran a controlled seven-phase study on MobileNetV2, EfficientNetV2 and eleven frozen pretrained backbones; a model scoring 0.965 AUC on dermoscopy sat near chance on clinical photographs, and adding darker-skin images alone did not help dark skin more than a matched lighter-skin control (3 seeds). Fine-tuning on clinical photographs raised AUC from 0.58 to 0.74, and an ensemble of frozen pretrained backbones reached 0.81 (nested five-fold cross-validation), showing the imaging domain had to match before representation helped.

**Refresh & Content Opportunity Scoring: Ranking Pages for Human Review Under Limited Capacity**
*FlyRank ML internship capstone research paper · nayab-khalid.github.io/FlyRank-AI-Internship · 2026*
Binary classification over 30K content pages from 32 pseudonymised clients, with the data contract verified against a 9.8M-row warehouse release and client-grouped validation so no client appeared in both training and test. The corrected model reached 0.596 Average Precision and 0.70 Precision@50, up from a 0.489 and 0.30 baseline.

## PROJECTS

**SkinSpectrum** | *TensorFlow/Keras, MobileNetV2, OpenCV, React Native, MongoDB, AWS | 2025 – Present*
Analyzes facial images for 9 skin conditions and melanoma risk using MobileNetV2 (ISIC + HAM10000); includes an AI assistant, product hub, progress tracking and community on a MongoDB Atlas + AWS backend.

**Netflix Dataset Analysis** | *Hadoop, HDFS, PySpark, Hive, HBase, Parquet, D3.js | 2026*
8,807 titles through HDFS → PySpark → Hive pipeline, visualised in a cross-filtering D3.js dashboard.

**Distributed Emotion Detector** | *Python, MPI (mpi4py), NLP, Flask | 2026*
Real-time emotion classification distributed across 5 MPI processes with a lexicon-based NLP engine, served via Flask.

**FocusFlow** | *React.js, Next.js, Node.js, Docker, Jenkins, Kubernetes, AWS | 2025*
Student productivity app deployed via Docker on AWS EC2, with a Jenkins CI/CD pipeline and Kubernetes orchestration.

**Hospital Management & Analytics** | *Power BI, Tableau, Star Schema | 2025*
Star-schema warehouse over 11,400+ records; Power BI and Tableau dashboards tracking revenue, workload and treatment patterns.

## TECHNICAL SKILLS

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|---|---|
| **Programming** | Python, Java, R |
| **Machine Learning & AI** | Scikit-learn, TensorFlow/Keras, NLP, Computer Vision |
| **Data Science** | Pandas, NumPy, Jupyter Notebook, Matplotlib, Seaborn, Power BI, Tableau |
| **Cloud & DevOps** | AWS, Docker, Jenkins, Git, GitHub |
| **Web Development** | React, Next.js, Node.js |
| **Databases & Big Data** | SQL, MongoDB, Hadoop, Spark, Hive |

## CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification — *Stanford University, 2026*
- Probability & Statistics for Machine Learning & Data Science — *DeepLearning.AI, 2026*
- Applied Social Network Analysis in Python — *University of Michigan, 2026*
- AI Fluency: Framework & Foundations — *Anthropic, 2026*
- Share Data Through the Art of Visualization — *Google, 2026*
- BCG Data Science Job Simulation — *Forage, 2026*
- Databases and SQL for Data Science — *IBM, 2025*
- CCNA: Introduction to Networks — *Cisco, 2024*

## AWARDS & VOLUNTEERING

- Merit Certificates — 2nd Position in Class, GPA 3.80 (2023) · 3rd Position in Class, GPA 3.71 (2025) · 1st Position in Class, GPA 3.93 (2026)
- Media Volunteer, Photography & Videography — COMSATS University Islamabad (2024)
- Registered Volunteer, Alkhidmat Foundation — Community Service (2026)

## LANGUAGES & INTERESTS

**Languages:** Urdu — Native · English — Professional working proficiency
**Interests:** Photography, Reading, Cinema, Music, Editing, Journaling